

Tong QFT — Problem Sheet 3

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Weyl and Dirac Representations of the Clifford Algebra, Q1

The Weyl representation of the Clifford algebra is given by

$$\gamma^0 = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} \quad (1.1)$$

Show that these indeed satisfy $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$, where $\mathbf{1}$ comes with an implicit 4×4 unit matrix. Find a unitary matrix U such that $(\gamma')^\mu = U\gamma^\mu U^\dagger$, where $(\gamma')^\mu$ form the Dirac representation of the Clifford algebra

$$(\gamma')^0 = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix}, \quad (\gamma')^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}. \quad (1.2)$$

Solution. Write $\sigma^\mu = (\mathbf{1}, \sigma)$, and $\bar{\sigma}^\mu = (\mathbf{1}, -\sigma)$. The anticommutator is given by

$$\begin{aligned} \{\gamma^\mu, \gamma^\nu\} &= \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma^\nu \\ \bar{\sigma}^\nu & 0 \end{pmatrix} + \begin{pmatrix} 0 & \sigma^\nu \\ \bar{\sigma}^\nu & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} \\ &= \begin{pmatrix} \sigma^\mu \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\mu & 0 \\ 0 & \bar{\sigma}^\mu \sigma^\nu + \bar{\sigma}^\nu \sigma^\mu \end{pmatrix} \\ &= 2\eta^{\mu\nu} \mathbf{1}. \end{aligned} \quad (1.3)$$

U diagonalise γ^0 , and left the rest unchanged. Therefore, we find

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{1} & \mathbf{1} \\ \mathbf{1} & -\mathbf{1} \end{pmatrix}. \quad (1.4)$$

The Lorentz Algebra from Gamma Matrices, Q2

Show that if $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$, then

$$[\gamma^\kappa \gamma^\lambda, \gamma^\mu \gamma^\nu] = 2\eta^{\lambda\mu} \gamma^\kappa \gamma^\nu - 2\eta^{\kappa\mu} \gamma^\lambda \gamma^\nu + 2\eta^{\lambda\nu} \gamma^\mu \gamma^\kappa - 2\eta^{\kappa\nu} \gamma^\mu \gamma^\lambda. \quad (2.1)$$

Show further that $S^{\mu\nu} \equiv \frac{1}{4}[\gamma^\mu, \gamma^\nu] = \frac{1}{2}(\gamma^\mu \gamma^\nu - \eta^{\mu\nu})$. Use this to confirm that the matrices $S^{\mu\nu}$ form a representation of the Lie algebra of the Lorentz group.

Solution. Note that

$$[\gamma^\kappa \gamma^\lambda, \gamma^\mu \gamma^\nu] = \gamma^\kappa \gamma^\lambda \gamma^\mu \gamma^\nu - \gamma^\mu \gamma^\nu \gamma^\kappa \gamma^\lambda. \quad (2.2)$$

Then, Hence, we find

$$\begin{aligned} \gamma^\kappa \gamma^\lambda \gamma^\mu \gamma^\nu &= 2\eta^{\lambda\mu} \gamma^\kappa \gamma^\nu - \gamma^\kappa \gamma^\mu \gamma^\lambda \gamma^\nu \\ &= 2\eta^{\lambda\mu} \gamma^\kappa \gamma^\nu - 2\eta^{\lambda\nu} \gamma^\kappa \gamma^\mu + \gamma^\kappa \gamma^\mu \gamma^\nu \gamma^\lambda \\ &= 2\eta^{\lambda\mu} \gamma^\kappa \gamma^\nu - 2\eta^{\lambda\nu} \gamma^\kappa \gamma^\mu + 2\eta^{\kappa\mu} \gamma^\nu \gamma^\lambda - 2\eta^{\lambda\nu} \gamma^\mu \gamma^\lambda + \gamma^\mu \gamma^\nu \gamma^\kappa \gamma^\lambda. \end{aligned} \quad (2.3)$$

Hence, we find

$$[\gamma^\kappa \gamma^\lambda, \gamma^\mu \gamma^\nu] = 2\eta^{\lambda\mu} \gamma^\kappa \gamma^\nu - 2\eta^{\kappa\mu} \gamma^\lambda \gamma^\nu + 2\eta^{\lambda\nu} \gamma^\mu \gamma^\kappa - 2\eta^{\kappa\nu} \gamma^\mu \gamma^\lambda. \quad (2.4)$$

Similarly,

$$\begin{aligned} S^{\mu\nu} &= \frac{1}{4}(\gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu) \\ &= \frac{1}{4}(2\gamma^\mu \gamma^\nu - 2\eta^{\mu\nu}) \\ &= \frac{1}{2}(\gamma^\mu \gamma^\nu - \eta^{\mu\nu}). \end{aligned} \quad (2.5)$$

The Lie algebra is given by

$$\begin{aligned} &= \frac{1}{4}(\gamma^\kappa \gamma^\lambda \gamma^\mu \gamma^\nu - \eta^{\kappa\lambda} \gamma^\mu \gamma^\nu - \eta^{\mu\nu} \gamma^\kappa \gamma^\lambda - \gamma^\mu \gamma^\nu \gamma^\kappa \gamma^\lambda + \eta^{\kappa\lambda} \gamma^\mu \gamma^\nu + \eta^{\mu\nu} \gamma^\kappa \gamma^\lambda) \\ &= \frac{1}{4}[\gamma^\kappa \gamma^\lambda, \gamma^\mu \gamma^\nu] \\ &= \frac{1}{4}(2\eta^{\lambda\mu} \gamma^\kappa \gamma^\nu - 2\eta^{\kappa\mu} \gamma^\lambda \gamma^\nu + 2\eta^{\lambda\nu} \gamma^\mu \gamma^\kappa - 2\eta^{\kappa\nu} \gamma^\mu \gamma^\lambda) \\ &= \frac{1}{2}(\eta^{\lambda\mu} \gamma^\kappa \gamma^\nu - \eta^{\kappa\mu} \gamma^\lambda \gamma^\nu + \eta^{\lambda\nu} \gamma^\mu \gamma^\kappa - \eta^{\kappa\nu} \gamma^\mu \gamma^\lambda) \\ &= \eta^{\lambda\mu} (S^{\kappa\nu} + \frac{1}{2}\eta^{\kappa\nu}) - \eta^{\kappa\mu} (S^{\lambda\nu} + \frac{1}{2}\eta^{\lambda\nu}) + \eta^{\lambda\nu} (S^{\mu\kappa} + \frac{1}{2}\eta^{\mu\kappa}) - \eta^{\kappa\nu} (S^{\mu\lambda} + \frac{1}{2}\eta^{\mu\lambda}) \\ &= \eta^{\lambda\mu} S^{\kappa\nu} - \eta^{\kappa\mu} S^{\lambda\nu} + \eta^{\lambda\nu} S^{\mu\kappa} - \eta^{\kappa\nu} S^{\mu\lambda}. \end{aligned} \quad (2.6)$$

Trace Technology and Gamma Matrix Identities, Q3

Using just the algebra $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ (i.e. without resorting to a particular representation), and defining $\gamma^5 = -i\gamma^0\gamma^1\gamma^2\gamma^3$, $\not{p} = p_\mu \gamma^\mu$ and $S^{\mu\nu} = \frac{1}{4}[\gamma^\mu, \gamma^\nu]$, prove the following results: (Some useful tricks include the cyclicity of the trace, and inserting $(\gamma^5)^2 = 1$ into a trace).

$$1. \text{Tr } \gamma^\mu = 0$$

2. $\text{Tr}(\gamma^\mu \gamma^\nu) = 4\eta^{\mu\nu}$
3. $\text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho) = 0$
4. $(\gamma^5)^2 = 1$
5. $\text{Tr} \gamma^5 = 0$
6. $\not{p} \not{q} = 2p \cdot q - \not{q} \not{p} = p \cdot q + 2S^{\mu\nu} p_\mu q_\nu$
7. $\text{Tr}(\not{p} \not{q}) = 4p \cdot q$
8. $\text{Tr}(\not{p}_1 \cdots \not{p}_n) = 0$ if n is odd
9. $\text{Tr}(\not{p}_1 \not{p}_2 \not{p}_3 \not{p}_4) = 4[(p_1 \cdot p_2)(p_3 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3) - (p_1 \cdot p_3)(p_2 \cdot p_4)]$
10. $\text{Tr}(\gamma^5 \not{p}_1 \not{p}_2) = 0$
11. $\gamma_\mu \not{p} \gamma^\mu = -2 \not{p}$
12. $\gamma_\mu \not{p}_1 \not{p}_2 \gamma^\mu = 4p_1 \cdot p_2$
13. $\gamma_\mu \not{p}_1 \not{p}_2 \not{p}_3 \gamma^\mu = -2 \not{p}_3 \not{p}_2 \not{p}_1$
14. $\text{Tr}(\gamma^5 \not{p}_1 \not{p}_2 \not{p}_3 \not{p}_4) = 4i \epsilon_{\mu\nu\rho\sigma} p_1^\mu p_2^\nu p_3^\rho p_4^\sigma$

Solution. Let us consider spinors in d -dimensions.

1. Trivial.

$$2. \quad \text{Tr}(\gamma^\mu \gamma^\nu) = 2\eta^{\mu\nu} \text{Tr}(\mathbf{1}) - \text{Tr}(\gamma^\nu \gamma^\mu) \implies \text{Tr}(\gamma^\mu \gamma^\nu) = d\eta^{\mu\nu}. \quad (3.1)$$

$$3. \quad \text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho) = -\text{Tr}(\gamma^\nu \gamma^\mu \gamma^\rho) = \text{Tr}(\gamma^\nu \gamma^\rho \gamma^\mu) = -\text{Tr}(\gamma^\rho \gamma^\nu \gamma^\mu) = 0. \quad (3.2)$$

$$4. \quad (\gamma^5)^2 = -\gamma^0 \gamma^1 \gamma^2 \gamma^3 \gamma^0 \gamma^1 \gamma^2 \gamma^3 = -\gamma^1 \gamma^2 \gamma^3 \gamma^1 \gamma^2 \gamma^3 = -\gamma^2 \gamma^3 \gamma^2 \gamma^3 = \mathbf{1}. \quad (3.3)$$

5. Will turn out to be trivial using the general trace of for γ -matrices.

$$6. \quad \not{p} \not{q} = p_\mu q_\nu (2\eta^{\mu\nu} - \gamma^\nu \gamma^\mu) = 2p \cdot q - \not{q} \not{p} = p_\mu q_\nu (\eta^{\mu\nu} + 2S^{\mu\nu}) = p \cdot q + 2S^{\mu\nu} p_\mu q_\nu. \quad (3.4)$$

$$7. \quad \text{Tr}(\not{p} \not{q}) = d p \cdot q. \quad (3.5)$$

8. Follows from iii by induction.

$$\begin{aligned} 9. \quad & \text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\lambda) = 2(\eta^{\mu\nu} \text{Tr}(\gamma^\rho \gamma^\lambda) - \eta^{\mu\rho} \text{Tr}(\gamma^\nu \gamma^\lambda) + \eta^{\mu\lambda} \text{Tr}(\gamma^\nu \gamma^\rho)) - \text{Tr}(\gamma^\nu \gamma^\rho \gamma^\lambda \gamma^\mu) \\ & \implies \text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\lambda) = d(\eta^{\mu\nu} \eta^{\rho\lambda} - \eta^{\mu\rho} \eta^{\nu\lambda} + \eta^{\mu\lambda} \eta^{\nu\rho}) \\ & \implies \text{Tr}(\not{p}_1 \not{p}_2 \not{p}_3 \not{p}_4) = d[(p_1 \cdot p_2)(p_3 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3) - (p_1 \cdot p_3)(p_2 \cdot p_4)]. \end{aligned}$$

10. A cleaner argument: by anticommutation of γ^5 with all γ^μ , the trace is totally antisymmetric in (μ, ν) . The only available rank-2 tensor in 4d is $\eta^{\mu\nu}$, which is symmetric. Hence the trace must vanish:

$$\text{Tr}(\gamma^5 \not{p}_1 \not{p}_2) = 0. \quad (3.7)$$

11. Insert $\eta_{\mu\nu} \gamma^\mu \gamma^\nu = d \mathbf{1}$ and use $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$:

$$\gamma_\mu \not{p} \gamma^\mu = p_\nu \gamma_\mu \gamma^\nu \gamma^\mu = p_\nu (2\eta_\mu^\nu - \gamma^\nu \gamma_\mu) \gamma^\mu = 2 \not{p} - d \not{p} = (2 - d) \not{p} = -2 \not{p}. \quad (3.8)$$

12. Apply the previous result twice. Using $\gamma_\mu \gamma^\nu \gamma^\rho \gamma^\mu = 4\eta^{\nu\rho}$ (proved similarly):

$$\gamma_\mu \not{\epsilon}_1 \not{\epsilon}_2 \gamma^\mu = p_{1\nu} p_{2\rho} \gamma_\mu \gamma^\nu \gamma^\rho \gamma^\mu = 4p_{1\nu} p_{2\rho} \eta^{\nu\rho} = 4p_1 \cdot p_2. \quad (3.9)$$

13. Similarly $\gamma_\mu \gamma^\nu \gamma^\rho \gamma^\sigma \gamma^\mu = -2\gamma^\sigma \gamma^\rho \gamma^\nu$, hence

$$\gamma_\mu \not{\epsilon}_1 \not{\epsilon}_2 \not{\epsilon}_3 \gamma^\mu = -2 \not{\epsilon}_3 \not{\epsilon}_2 \not{\epsilon}_1. \quad (3.10)$$

14. By the same antisymmetry argument as in (x), the trace $\text{Tr}(\gamma^5 \not{\epsilon}_1 \not{\epsilon}_2 \not{\epsilon}_3 \not{\epsilon}_4)$ must be totally antisymmetric in the four momentum indices. The only such rank-4 tensor in 4d is $\epsilon^{\mu\nu\rho\sigma}$, so

$$\text{Tr}(\gamma^5 \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) = c \epsilon^{\mu\nu\rho\sigma}. \quad (3.11)$$

The constant c is fixed by setting $(\mu, \nu, \rho, \sigma) = (0, 1, 2, 3)$:

$$\text{Tr}(\gamma^5 \gamma^0 \gamma^1 \gamma^2 \gamma^3) = i \text{Tr}(\gamma^5 \gamma^5) = 4i, \quad (3.12)$$

using $\gamma^5 = -i\gamma^0 \gamma^1 \gamma^2 \gamma^3$ and $(\gamma^5)^2 = \mathbf{1}$. With $\epsilon^{0123} = -\epsilon_{0123} = -1$, we find $c = -4i$, giving

$$\text{Tr}(\gamma^5 \not{\epsilon}_1 \not{\epsilon}_2 \not{\epsilon}_3 \not{\epsilon}_4) = -4i \epsilon_{\mu\nu\rho\sigma} p_1^\mu p_2^\nu p_3^\rho p_4^\sigma. \quad (3.13)$$

Plane-Wave Spinor Solutions and Orthogonality, Q4

The plane-wave solutions to the Dirac equation are

$$u^s(\mathbf{p}) = \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^s \\ \sqrt{p \cdot \bar{\sigma}} \xi^s \end{pmatrix} \quad \text{and} \quad v^s(\mathbf{p}) = \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^s \\ -\sqrt{p \cdot \bar{\sigma}} \xi^s \end{pmatrix} \quad (4.1)$$

where $\sigma^\mu = (1, \sigma^i)$ and $\bar{\sigma}^\mu = (1, -\sigma^i)$ and ξ^s , with $s = 1, 2$, is a basis of orthonormal two-component spinors, satisfying $(\xi^r)^\dagger \cdot \xi^s = \delta^{rs}$. Show that

$$\begin{aligned} u^r(\mathbf{p})^\dagger \cdot u^s(\mathbf{p}) &= 2p_0 \delta^{rs} \\ \bar{u}^r(\mathbf{p}) \cdot u^s(\mathbf{p}) &= 2m \delta^{rs} \end{aligned} \quad (4.2)$$

and similarly,

$$\begin{aligned} v^r(\mathbf{p})^\dagger \cdot v^s(\mathbf{p}) &= 2p_0 \delta^{rs} \\ \bar{v}^r(\mathbf{p}) \cdot v^s(\mathbf{p}) &= -2m \delta^{rs} \end{aligned} \quad (4.3)$$

Show also that the orthonality condition between u and v is

$$\bar{u}^s(\mathbf{p}) \cdot v^r(\mathbf{p}) = 0 \quad (4.4)$$

while taking the inner product using $\dagger$ requires an extra minus sign

$$u^r(\mathbf{p})^\dagger \cdot v^s(-\mathbf{p}) = 0. \quad (4.5)$$

Solution.

$$\begin{aligned}
u^r(\mathbf{p})^\dagger \cdot u^s(\mathbf{p}) &= \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^{\dagger r} & \sqrt{p \cdot \bar{\sigma}} \xi^{\dagger r} \end{pmatrix} \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^s \\ \sqrt{p \cdot \bar{\sigma}} \xi^s \end{pmatrix} \\
&= \delta^{rs} \left[\sqrt{(p \cdot \sigma)(p \cdot \sigma^\dagger)} + \sqrt{(p \cdot \bar{\sigma})(p \cdot \bar{\sigma}^\dagger)} \right] \\
&= 2\delta^{rs} \sqrt{(p \cdot \sigma)(p \cdot \sigma^\dagger)},
\end{aligned} \tag{4.6}$$

Since the Pauli matrices are Hermitian, we find

$$u^r(\mathbf{p})^\dagger \cdot u^s(\mathbf{p}) = 2\delta^{rs}(p \cdot \sigma + p \cdot \bar{\sigma}) = 2p_0\delta^{rs}. \tag{4.7}$$

Similarly, we find

$$\begin{aligned}
\bar{u}^r(\mathbf{p}) \cdot u^s(\mathbf{p}) &= \begin{pmatrix} \sqrt{p \cdot \bar{\sigma}} \xi^{\dagger r} & \sqrt{p \cdot \sigma} \xi^{\dagger r} \end{pmatrix} \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^s \\ \sqrt{p \cdot \bar{\sigma}} \xi^s \end{pmatrix} \\
&= \delta^{rs} \left[\sqrt{(p \cdot \sigma)(p \cdot \bar{\sigma}^\dagger)} + \sqrt{(p \cdot \bar{\sigma})(p \cdot \sigma^\dagger)} \right] \\
&= 2\delta^{rs} \sqrt{(p \cdot \sigma)(p \cdot \bar{\sigma})} \\
&= 2m\delta^{rs},
\end{aligned} \tag{4.8}$$

where we have expanded $p \cdot \sigma$ and $p \cdot \bar{\sigma}$ and evaluated them explicitly.

Similarly, one can easily obtain

$$\begin{aligned}
v^r(\mathbf{p})^\dagger \cdot v^s(\mathbf{p}) &= 2p_0\delta^{rs} \\
\bar{v}^r(\mathbf{p}) \cdot v^s(\mathbf{p}) &= -2m\delta^{rs}.
\end{aligned} \tag{4.9}$$

Now, we demonstrate the orthogonality condition,

$$\bar{u}^s(\mathbf{p}) \cdot v^r(\mathbf{p}) = \begin{pmatrix} \sqrt{p \cdot \bar{\sigma}} \xi^{\dagger s} & \sqrt{p \cdot \sigma} \xi^{\dagger s} \end{pmatrix} \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^r \\ -\sqrt{p \cdot \bar{\sigma}} \xi^r \end{pmatrix} = 0, \tag{4.10}$$

by Hermiticity of Pauli matrices. Similarly,

$$u^r(\mathbf{p})^\dagger \cdot v^s(-\mathbf{p}) = \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^{\dagger r} & \sqrt{p \cdot \bar{\sigma}} \xi^{\dagger r} \end{pmatrix} \begin{pmatrix} \sqrt{\tilde{p} \cdot \sigma} \xi^s \\ -\sqrt{\tilde{p} \cdot \bar{\sigma}} \xi^s \end{pmatrix}, \tag{4.11}$$

where $\tilde{p}^\mu = (p_0, -\mathbf{p})$. Now, note that $\tilde{p} \cdot \sigma = p \cdot \bar{\sigma}$, we see that the above inner product vanishes.

Spin Sums and Completeness Relation, Q5

Using the same notation as Question 4 show that

$$\begin{aligned}
\sum_{s=1}^2 u^s(\mathbf{p}) \bar{u}^s(\mathbf{p}) &= \not{p} + m \\
\sum_{s=1}^2 v^s(\mathbf{p}) \bar{v}^s(\mathbf{p}) &= \not{p} - m
\end{aligned} \tag{5.1}$$

where, rather than being contracted, the two spinors on the left-hand side are placed back to back to form a 4×4 matrix.

Solution.

$$\begin{aligned}
\sum_{s=1}^2 u^s(\mathbf{p}) \bar{u}^s(\mathbf{p}) &= \sum_{s=1}^2 \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^s \\ \sqrt{p \cdot \bar{\sigma}} \xi^s \end{pmatrix} (\sqrt{p \cdot \bar{\sigma}} \xi^{\dagger s} \quad \sqrt{p \cdot \sigma} \xi^{\dagger s}) \\
&= \begin{pmatrix} m & p \cdot \sigma \\ p \cdot \bar{\sigma} & m \end{pmatrix} \sum_{s=1}^2 \xi^s \xi^{\dagger s}.
\end{aligned} \tag{5.2}$$

The latter sum can be obtained by considering

$$\sum_{s=1}^2 \delta^{sa} \delta^{sb} = \delta^{ab} \implies \sum_{s=1}^2 (\xi^{\dagger s} \cdot \xi^a) (\xi^{\dagger b} \cdot \xi^s) = \xi^{\dagger b} \cdot \xi^a \implies \sum_{s=1}^2 \xi_a^s \xi_b^{\dagger s} = \delta_{ab}. \tag{5.3}$$

Therefore, we find

$$\sum_{s=1}^2 u^s(\mathbf{p}) \bar{u}^s(\mathbf{p}) = \not{p} + m. \tag{5.4}$$

Almost identically, we find

$$\sum_{s=1}^2 v^s(\mathbf{p}) \bar{v}^s(\mathbf{p}) = \not{p} - m. \tag{5.5}$$

Canonical Quantization of the Dirac Field, Q6

The Fourier decomposition of the Dirac operator $\psi(\mathbf{x})$ and the conjugate field $\psi^\dagger(\mathbf{x})$ is given by

$$\begin{aligned}
\psi(\mathbf{x}) &= \sum_{s=1}^2 \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} [b_{\mathbf{p}}^s u^s(\mathbf{p}) e^{+i\mathbf{p} \cdot \mathbf{x}} + c_{\mathbf{p}}^{s\dagger} v^s(\mathbf{p}) e^{-i\mathbf{p} \cdot \mathbf{x}}] \\
\psi^\dagger(\mathbf{x}) &= \sum_{s=1}^2 \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} [b_{\mathbf{p}}^{s\dagger} u^s(\mathbf{p})^\dagger e^{-i\mathbf{p} \cdot \mathbf{x}} + c_{\mathbf{p}}^s v^s(\mathbf{p})^\dagger e^{+i\mathbf{p} \cdot \mathbf{x}}]
\end{aligned} \tag{6.1}$$

The creation and annihilation operators are taken to satisfy

$$\begin{aligned}
\{b_{\mathbf{p}}^r, b_{\mathbf{q}}^{s\dagger}\} &= (2\pi)^3 \delta^{rs} \delta^{(3)}(\mathbf{p} - \mathbf{q}) \\
\{c_{\mathbf{p}}^r, c_{\mathbf{q}}^{s\dagger}\} &= (2\pi)^3 \delta^{rs} \delta^{(3)}(\mathbf{p} - \mathbf{q})
\end{aligned} \tag{6.2}$$

with all other anti-commutators vanishing,

$$\{b_{\mathbf{p}}^r, b_{\mathbf{q}}^s\} = \{c_{\mathbf{p}}^r, c_{\mathbf{q}}^s\} = \{b_{\mathbf{p}}^r, c_{\mathbf{q}}^{s\dagger}\} = \{b_{\mathbf{p}}^r, c_{\mathbf{q}}^s\} = \dots = 0. \quad (6.3)$$

Show that these imply that the field and its conjugate momenta satisfy the anti-commutation relations,

$$\begin{aligned} \{\psi_{\alpha}(\mathbf{x}), \psi_{\beta}(\mathbf{y})\} &= \{\psi_{\alpha}^{\dagger}(\mathbf{x}), \psi_{\beta}^{\dagger}(\mathbf{y})\} = 0 \\ \{\psi_{\alpha}(\mathbf{x}), \psi_{\beta}^{\dagger}(\mathbf{y})\} &= \delta_{\alpha\beta} \delta^{(3)}(\mathbf{x} - \mathbf{y}). \end{aligned} \quad (6.4)$$

(Note: The calculation is very similar to that for the bosonic field, but at some point you will need to make use of the identities (5.1) and (vouter)).

Solution. We compute $\{\psi_{\alpha}(\mathbf{x}), \psi_{\beta}^{\dagger}(\mathbf{y})\}$ directly. Substituting the Fourier decompositions,

$$\begin{aligned} \{\psi_{\alpha}(\mathbf{x}), \psi_{\beta}^{\dagger}(\mathbf{y})\} &= \sum_{r,s} \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \int \frac{d^3q}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{q}}}} \\ &\times \left[\{b_{\mathbf{p}}^r, b_{\mathbf{q}}^{s\dagger}\} u_{\alpha}^r(\mathbf{p}) u_{\beta}^{s\dagger}(\mathbf{q}) e^{i\mathbf{p}\cdot\mathbf{x} - i\mathbf{q}\cdot\mathbf{y}} \right. \\ &\quad \left. + \{c_{\mathbf{p}}^{r\dagger}, c_{\mathbf{q}}^s\} v_{\alpha}^r(\mathbf{p}) v_{\beta}^{s\dagger}(\mathbf{q}) e^{-i\mathbf{p}\cdot\mathbf{x} + i\mathbf{q}\cdot\mathbf{y}} \right] \end{aligned} \quad (6.5)$$

where all cross terms $\{b, c\}$ and $\{b, c^{\dagger}\}$ vanish. Then,

$$\{\psi_{\alpha}(\mathbf{x}), \psi_{\beta}^{\dagger}(\mathbf{y})\} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} \sum_s \left[u_{\alpha}^s(\mathbf{p}) u_{\beta}^{s\dagger}(\mathbf{p}) e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} + v_{\alpha}^s(\mathbf{p}) v_{\beta}^{s\dagger}(\mathbf{p}) e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \right]. \quad (6.6)$$

Now we use the spin sum identities. Note that $\bar{u} = u^{\dagger} \gamma^0$ so $u^{\dagger} = \bar{u} \gamma^0$, giving

$$\sum_s u^s u^{s\dagger} = \sum_s u^s \bar{u}^s \gamma^0 = (\not{p} + m) \gamma^0, \quad \sum_s v^s v^{s\dagger} = (\not{p} - m) \gamma^0. \quad (6.7)$$

Substituting back,

$$\{\psi_{\alpha}(\mathbf{x}), \psi_{\beta}^{\dagger}(\mathbf{y})\} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} \left[(\not{p} + m)_{\alpha\dot{\beta}} (\gamma^0)_{\dot{\beta}\beta} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} + (\not{p} - m)_{\alpha\dot{\beta}} (\gamma^0)_{\dot{\beta}\beta} e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \right]. \quad (6.8)$$

Expanding $\not{p} = p_0 \gamma^0 - p_i \gamma^i$ and using $(\gamma^0)^2 = \mathbf{1}$, the m terms cancel between the two exponentials after relabelling $\mathbf{p} \rightarrow -\mathbf{p}$ in the second integral, and the $p_i \gamma^i \gamma^0$ terms also cancel by the same relabelling. What remains is

$$\begin{aligned} \{\psi_{\alpha}(\mathbf{x}), \psi_{\beta}^{\dagger}(\mathbf{y})\} &= \int \frac{d^3p}{(2\pi)^3} \frac{2p_0}{2E_{\mathbf{p}}} (\gamma^0 \gamma^0)_{\alpha\beta} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \\ &= \delta_{\alpha\beta} \int \frac{d^3p}{(2\pi)^3} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} = \delta_{\alpha\beta} \delta^{(3)}(\mathbf{x} - \mathbf{y}). \end{aligned} \quad (6.9)$$

The vanishing of $\{\psi_{\alpha}(\mathbf{x}), \psi_{\beta}(\mathbf{y})\}$ follows similarly, since the cross terms $\{b_{\mathbf{p}}^r, c_{\mathbf{q}}^{s\dagger}\} = 0$ and the surviving terms $\{b, b\} = \{c^{\dagger}, c^{\dagger}\} = 0$ all vanish.

The Dirac Hamiltonian, Q7

Using the results of Question 6, show that the quantum Hamiltonian

$$H = \int d^3x \bar{\psi}(-i\gamma^i \partial_i + m)\psi \quad (7.1)$$

can be written, after normal ordering, as

$$H = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \sum_{s=1}^2 [b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s + c_{\mathbf{p}}^{s\dagger} c_{\mathbf{p}}^s]. \quad (7.2)$$

(Note: Again, the calculation is very similar to that for the bosonic field. This time you will need to make use of the identities derived in Questions 4 and 5).

Solution. Substituting the Fourier decompositions and using $\bar{\psi} = \psi^\dagger \gamma^0$,

$$H = \sum_{r,s} \int d^3x \int \frac{d^3p}{(2\pi)^3} \frac{E_{\mathbf{p}}}{\sqrt{2E_{\mathbf{p}}}} \int \frac{d^3q}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{q}}}} \quad (7.3)$$

$$\times [b_{\mathbf{q}}^{r\dagger} u^r(\mathbf{q})^\dagger e^{-i\mathbf{q}\cdot\mathbf{x}} + c_{\mathbf{q}}^r v^r(\mathbf{q})^\dagger e^{+i\mathbf{q}\cdot\mathbf{x}}] [b_{\mathbf{p}}^s u^s(\mathbf{p}) e^{+i\mathbf{p}\cdot\mathbf{x}} - c_{\mathbf{p}}^{s\dagger} v^s(\mathbf{p}) e^{-i\mathbf{p}\cdot\mathbf{x}}]$$

where we applied $(-i\gamma^i \partial_i + m)u^s(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{x}} = E_{\mathbf{p}}\gamma^0 u^s e^{i\mathbf{p}\cdot\mathbf{x}}$ and $(-i\gamma^i \partial_i + m)v^s e^{-i\mathbf{p}\cdot\mathbf{x}} = -E_{\mathbf{p}}\gamma^0 v^s e^{-i\mathbf{p}\cdot\mathbf{x}}$ from the Dirac equation, and used $(\gamma^0)^2 = \mathbf{1}$ to cancel the γ^0 from $\bar{\psi} = \psi^\dagger \gamma^0$. Integrating over d^3x gives delta functions, setting $\mathbf{q} = \mathbf{p}$ for diagonal terms and $\mathbf{q} = -\mathbf{p}$ for cross terms:

$$H = \sum_{r,s} \int \frac{d^3p}{(2\pi)^3} \frac{E_{\mathbf{p}}}{2E_{\mathbf{p}}} \left[u^r(\mathbf{p})^\dagger u^s(\mathbf{p}) b_{\mathbf{p}}^{r\dagger} b_{\mathbf{p}}^s - v^r(\mathbf{p})^\dagger v^s(\mathbf{p}) c_{\mathbf{p}}^r c_{\mathbf{p}}^{s\dagger} \right. \quad (7.4)$$

$$\left. + u^r(-\mathbf{p})^\dagger (-v^s(\mathbf{p})) b_{-\mathbf{p}}^{r\dagger} c_{\mathbf{p}}^{s\dagger} + v^r(-\mathbf{p})^\dagger u^s(\mathbf{p}) c_{-\mathbf{p}}^r b_{\mathbf{p}}^s \right].$$

The cross terms vanish by the orthogonality relation $u^r(-\mathbf{p})^\dagger v^s(\mathbf{p}) = 0$. Applying the normalisation $u^r(\mathbf{p})^\dagger u^s(\mathbf{p}) = v^r(\mathbf{p})^\dagger v^s(\mathbf{p}) = 2E_{\mathbf{p}}\delta^{rs}$ from (5) and (6),

$$H = \sum_s \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} [b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s - c_{\mathbf{p}}^s c_{\mathbf{p}}^{s\dagger}] \quad (7.5)$$

$$= \sum_s \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \left[b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s - \left(-c_{\mathbf{p}}^{s\dagger} c_{\mathbf{p}}^s + (2\pi)^3 \delta^{(3)}(0) \right) \right].$$

After normal ordering, the vacuum contribution $\delta^{(3)}(0)$ is discarded and the sign of the c term flips, giving

$$H = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \sum_{s=1}^2 [b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s + c_{\mathbf{p}}^{s\dagger} c_{\mathbf{p}}^s]. \quad (7.6)$$

Note the crucial sign difference from the bosonic case: here normal ordering the anticommuting $c_{\mathbf{p}}^s c_{\mathbf{p}}^{s\dagger}$ flips the sign to give a positive contribution, whereas for bosons it produces an infinite addi-

tive constant.

The Spin-Statistics Theorem, Q8

The purpose of this question is to give you a glimpse into the spin-statistics theorem. This theorem roughly says that if you try to quantize a field with the wrong statistics, bad things will happen. Here we'll see what goes wrong if you try to quantize a spin 1/2 field as a boson. We start with the usual decomposition (11). This time we choose bosonic commutation relations for the annihilation and creation operators,

$$\begin{aligned} &= (2\pi)^3 \delta^{rs} \delta^{(3)}(\mathbf{p} - \mathbf{q}) \\ [c_{\mathbf{p}}^r, c_{\mathbf{q}}^{s\dagger}] &= -(2\pi)^3 \delta^{rs} \delta^{(3)}(\mathbf{p} - \mathbf{q}) \end{aligned} \quad (8.1)$$

with all other commutators vanishing. Note the strange minus sign for the c operators. Repeat the calculation of Question 6 to show that these are equivalent to the commutation relations,

$$\begin{aligned} &= [\psi_{\alpha}^{\dagger}(\mathbf{x}), \psi_{\beta}^{\dagger}(\mathbf{y})] = 0 \\ [\psi_{\alpha}(\mathbf{x}), \psi_{\beta}^{\dagger}(\mathbf{y})] &= \delta_{\alpha\beta} \delta^{(3)}(\mathbf{x} - \mathbf{y}). \end{aligned} \quad (8.2)$$

Now repeat the calculation of Question 7, to show that, after normal ordering, the Hamiltonian is given by

$$H = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \sum_{s=1}^2 [b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s - c_{\mathbf{p}}^{s\dagger} c_{\mathbf{p}}^s]. \quad (8.3)$$

This Hamiltonian is not bounded below: you can lower the energy indefinitely by creating more and more c particles. This is the reason a theory of bosonic spin 1/2 particles is sick.

Solution. The calculation proceeds exactly as in Questions 6 and 7. The field commutator gives

$$\begin{aligned} &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} \sum_s [u_{\alpha}^s(\mathbf{p}) u_{\beta}^{s\dagger}(\mathbf{p}) e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})} - v_{\alpha}^s(\mathbf{p}) v_{\beta}^{s\dagger}(\mathbf{p}) e^{-i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})}] \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} [(\not{\mathbf{p}} + m) \gamma^0 e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})} - (\not{\mathbf{p}} - m) \gamma^0 e^{-i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})}]_{\alpha\beta} \\ &= \delta_{\alpha\beta} \delta^{(3)}(\mathbf{x} - \mathbf{y}) \end{aligned} \quad (8.4)$$

where the minus sign from $[c, c^{\dagger}] = -(2\pi)^3 \delta^{(3)}$ cancels the minus sign in the $v\bar{v}$ spin sum, reproducing the same result as the fermionic case. For the Hamiltonian,

$$\begin{aligned} H &= \sum_s \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} [b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s + c_{\mathbf{p}}^s c_{\mathbf{p}}^{s\dagger}] \\ &= \sum_s \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} [b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s + c_{\mathbf{p}}^{s\dagger} c_{\mathbf{p}}^s + (2\pi)^3 \delta^{(3)}(0)]. \end{aligned} \quad (8.5)$$

Note that now $c_{\mathbf{p}}^s c_{\mathbf{p}}^{s\dagger} = c_{\mathbf{p}}^{s\dagger} c_{\mathbf{p}}^s + (2\pi)^3 \delta^{(3)}(0)$ since these are bosonic operators. After normal ordering,

$$H = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \sum_{s=1}^2 [b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s - c_{\mathbf{p}}^{s\dagger} c_{\mathbf{p}}^s] \quad (8.6)$$

where the minus sign arises from the $[c, c^\dagger] = -(2\pi)^3 \delta^{(3)}$ relation flipping the sign under normal ordering.